

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Software Quality Assurance & Testing (CS-468)
Assignment # 2 Solution

Semester: VIII
Announced Date: April 15, 2026
Total Marks: 3
Instructor Name: Farhana Ashraf

Batch: 2022F
Due Date: April 23, 2026 for all sections
Marks Obtained:

- Instructions:**
- Attempt all parts of the question.
 - All parts carry equal parts.
 - Submit your **handwritten** assignment in form of hardcopy **in person**. Use **A4** size paper to solve the assignment.

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 2	Write and Use test cases to evaluate functional and non-functional specification requirements for various test types /levels / techniques and types of information systems and technologies.	PLO_4 (Design/ Development of Solutions)	C3 (Applying)

Q1. [3 mark]

- (a) A mail-order company selling flower seeds charges £3.95 for postage and packing on all orders up to £20 value and £4.95 for orders above £20 value and up to £40 value. For orders above £40 value there is no charge for postage and packing. If you were using equivalence partitioning to write test cases for the postage and packing charges what valid and invalid partitions would you **discover**? **Write** test cases using boundary value analysis.

- Answer:**
- Assumptions:**
1. Considering values in pounds only. No pence values are being considered.
 2. Minimum order could be of £1
 3. Maximum order could be of £1000

Partition	Type
{-∞, , -1}	Invalid
{0}	Invalid
{1, , 20}	Valid
{21, , 40}	Valid
{41, , 1000}	Valid
{1001, , ∞}	Invalid

To find sequence:
{-∞, , -1} {0} {1, , 20} {21, , 40} {41, , 1000} {1001, , ∞}

To find boundary values for each partition:

Left Neighbor	Partition	Right Neighbors	Boundary values	Test cases Required
N/A	{-∞, , -1}	{0}	{-1,0}	2
{-∞, , -1}	{0}	{1, , 20}	{-1, 0, 1}	3
{0}	{1, , 20}	{21, , 40}	{0, 1, 20, 21}	4
{1, , 20}	{21, , 40}	{41, , 1000}	{20, 21, 40, 41}	4
{21, , 40}	{41, , 1000}	{1001, , ∞}	{40, 41, 1000, 1001}	4

{41, , 1000}	{1001, , ∞}	N/A	{1000, 1001}	2
--------------------	-------------------	-----	--------------	---

Test cases using boundary value analysis:

S. No.	Partition	Test Data – Boundary values (Orders in £)	Expected Outcome (Postage charges in £)
1	{-∞, , -1}	-1	Error – Invalid Input
2		0	Error – Invalid Input
3	{0}	-1	Error – Invalid Input
4		0	Error – Invalid Input
5		1	£3.95
6	{1, , 20}	0	Error – Invalid Input
7		1	£3.95
8		20	£3.95
9		21	£4.95
10		20	£3.95
11	{21, , 40}	21	£4.95
12		40	£4.95
13		41	£0 (Free)
14		40	£4.95
15	{41, , 1000}	41	£0 (Free)
16		1000	£0 (Free)
17		1001	Error – Invalid Input
18		1000	£0 (Free)
19	{1001, , ∞}	1001	Error – Invalid Input

(b) A courier company provide its services across the country only. They do not offer international shipments. For gold card members, transactions over 80 dollars are entitled for free delivery provided that the parcel weight is below 10kg. **Construct** decision tree and decision table to **identify** the business rules. Also **write** positive test cases based on the identified business rules.

Answer:

Conditions:

- International shipment?
- Gold card member?
- Transaction over 80 dollars?
- Parcel weight is below 10kg?

Actions:

- Can avail free delivery

Total No. of conditions= 4 => **Total no. of business rules = 2⁴ = 16**

Decision Table:

Step 1:

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16
Conditions																
International shipment?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
Gold card member?	T	T	T	T	F	F	F	F	T	T	T	T	F	F	F	F
Transaction over 80 dollars?	T	T	F	F	T	T	F	F	T	T	F	F	T	T	F	F
Parcel weight is below 10kg	T	F	T	F	T	F	T	F	T	F	T	F	T	F	T	F
Actions																
Can avail free delivery	F	F	F	F	F	F	F	F	T	F	F	F	F	F	F	F

Step 2: Finding don't care conditions

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16
Conditions																
International shipment?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
Gold card member?	X	X	X	X	X	X	X	X	T	T	T	T	F	F	F	F
Transaction over 80 dollars?	X	X	X	X	X	X	X	X	T	T	F	F	X	X	X	X
Parcel weight is below 10kg	X	X	X	X	X	X	X	X	T	F	X	X	X	X	X	X
Actions																
Can avail free delivery	F	F	F	F	F	F	F	F	T	F	F	F	F	F	F	F

Step 3: Eliminating redundant business rules.

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16
Conditions																
International shipment?	T	T	T	T	T	T	T	T	F	F	F	F	F	F	F	F
Gold card member?	X	X	X	X	X	X	X	X	T	T	T	T	F	F	F	F
Transaction over 80 dollars?	X	X	X	X	X	X	X	X	T	T	F	F	X	X	X	X
Parcel weight is below 10kg	X	X	X	X	X	X	X	X	T	F	X	X	X	X	X	X
Actions																
Can avail free delivery	F	F	F	F	F	F	F	F	T	F	F	F	F	F	F	F

There are 5 business rules after eliminating duplicate entries.

	R1	R2	R3	R4	R5
Conditions					
International shipment?	T	F	F	F	F
Gold card member?	X	T	T	T	F
Transaction over 80 dollars?	X	T	T	F	X
Parcel weight is below 10kg	X	T	F	X	X
Actions					
Can avail free delivery	F	T	F	F	F

Positive Test cases for identified business rules:

S. No.	Test Data						Expected Outcome
	Destination	Payment type	Member	Membership Type	Amount	Parcel weight	
1	Canada	Credit Card	No	N/A	\$80	11kg	Don't offer services
2	Karachi	CoD	Yes	Gold	\$81	9kg	No shipping Charges
3	Lahore	Debit card	Yes	Gold	\$100	10kg	Has to pay shipping charges
4	Islamabad	Easy Paisa	Yes	Gold	\$100	10kg	Has to pay shipping charges
5	Hyderabad	Jazz Cash	Yes	Gold	\$100	10kg	Has to pay shipping charges

Decision Tree:

